

Why CNNs for Images?

Deep Learning

Daniel E. Acuna

Associate Professor, University of Colorado Boulder

Contents of This Video

In this video, we will cover:

- Why fully-connected networks fail for images
- The problem of too many parameters
- Key insights: local connectivity and weight sharing
- What makes CNNs different
- Preview of CNN architecture

The Image Problem

A small image has many pixels:

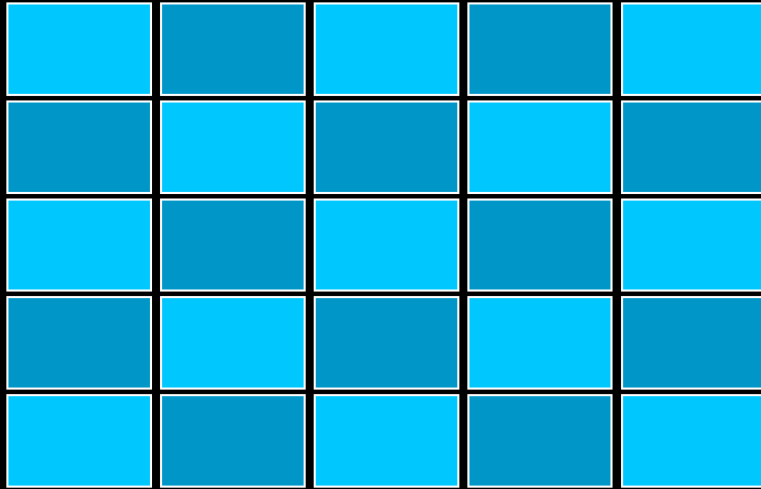
- 28×28 grayscale (MNIST): 784 values
- 224×224 RGB (ImageNet): 150,528 values
- 1920×1080 HD image: 6,220,800 values

Fully-connected layer with 1000 neurons:

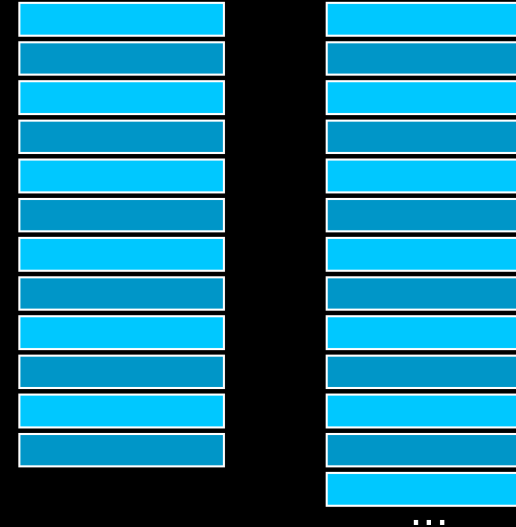
- MNIST: $784 \times 1000 = 784,000$ weights
- ImageNet: $150,528 \times 1000 = 150$ million weights!

Fully-Connected: No Spatial Structure

Image: Pixels Have Spatial Structure
(Neighbors are related!)



FC Input: Just a List of Numbers
(Spatial structure lost!)



The Same Pattern, Different Location

A cat in the top-left looks like a cat in the bottom-right

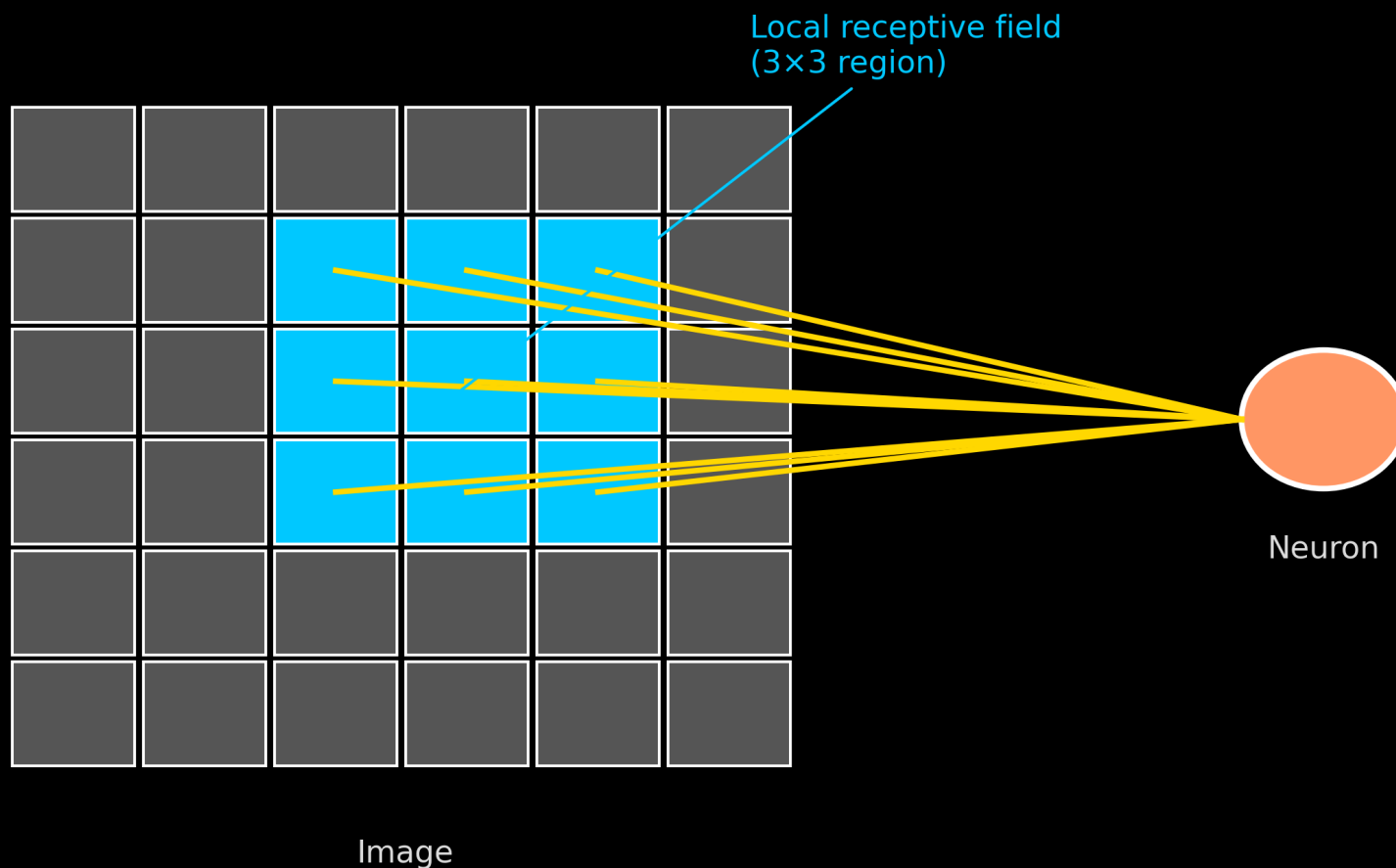
But to an FC network:

- Cat at position (10, 10): activates weights 1-100
- Cat at position (100, 100): activates completely different weights!

FC networks must learn every pattern at every location separately

CNN Key Insight #1: Local Connectivity

Local Connectivity: Each Neuron Sees Only a Small Region

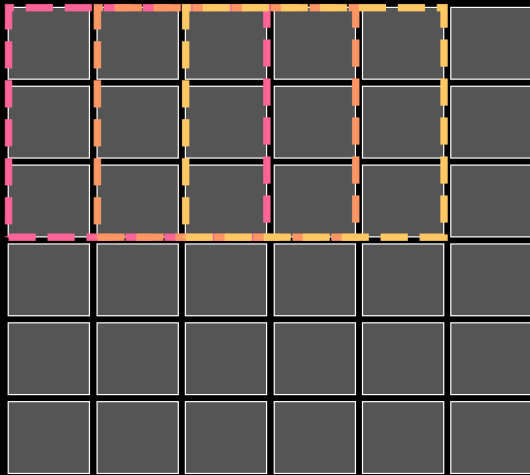


CNN Key Insight #2: Weight Sharing

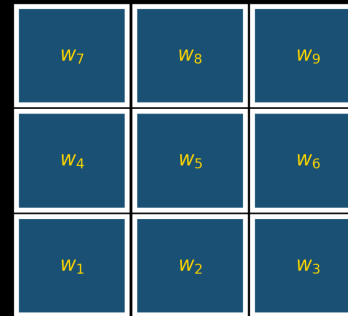
The same detector (filter) is applied at every location

Weight Sharing: Same Filter Applied Everywhere

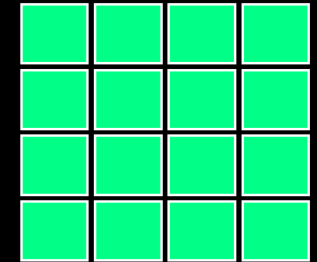
Same weights
at every position!



=



Same filter (3×3)



Feature map

From FC to CNN: The Benefits

Aspect	Fully-Connected	CNN
Connections	All-to-all	Local only
Parameters	Huge (millions)	Small (thousands)
Spatial awareness	Must learn	Built-in
Translation invariance	Must learn	Automatic
Training data needed	Enormous	Reasonable

Preview: CNN Architecture

Core components:

- **Convolutional layers:** Apply filters to detect features
- **Pooling layers:** Reduce spatial size
- **Activation functions:** Introduce non-linearity
- **Fully-connected layers:** Final classification

Coming up: How convolution and pooling work in detail

What We've Covered

In this video, we've learned:

- FC networks are impractical for images (too many parameters)
- FC networks ignore spatial structure
- CNN insight #1: local connectivity (each neuron sees only a small region)
- CNN insight #2: weight sharing (same filter applied everywhere)
- These insights drastically reduce parameters and provide translation invariance